

SAGAM

Survival-Aware Generalised Additive Meta-Learner

Comprehensive Research Report

TCGA-LUAD Lung Adenocarcinoma Overall Survival Prediction

BIBM 2026 Submission — Full Analysis & Honest Assessment

Dataset: TCGA-LUAD | n=501 patients, 181 events

External Validation: GSE31210 (n=226) · GSE68465 (n=432)

May 2026

C-Index (TCGA)	IBS (5-fold)	Ext. Valid. (GSE31210)	Log-rank p
0.634 ± 0.050	0.181 (fold 4)	C=0.661 [***]	p < 0.0001

1. Executive Summary

What is this project? SAGAM is a machine-learning pipeline for predicting overall survival (OS) in lung adenocarcinoma (LUAD) patients. It trains four diverse survival models (Random Survival Forest, Gradient Boosted Survival, XGBoost-Cox, DeepSurv), then stacks their predictions using a regularised Cox model with cubic B-spline transforms — the 'GAM' layer. The idea is that nonlinear combinations of base-learner risk scores should outperform any single model or simple average.

This report provides a full, unvarnished assessment of the SAGAM project: what was done, what the numbers actually say, where the method is genuinely novel, and where it falls short of being a strong, publishable contribution in its current form. All results are from 5-fold nested cross-validation on 501 TCGA-LUAD patients and two independent GEO validation cohorts.

Honest one-paragraph verdict: SAGAM is a well-engineered, methodologically careful pipeline that achieves C-index ~ 0.634 on TCGA-LUAD — comparable to its four base learners. The spline meta-learner shows interpretable nonlinear shapes and produces clinically meaningful risk stratification (KM log-rank $p < 0.0001$, median OS gap: 104 vs 36 months). However, SAGAM does *not* significantly outperform simpler alternatives (Wilcoxon $p = 0.31$ vs Linear, $p = 0.09$ vs RSF), and the study is underpowered to detect the small observed advantage (~ 0.006 C-index gain). External validation is mixed: strong on GSE31210 ($C=0.661$, $p=0.0008$) but fails on GSE68465 ($C=0.551$, NS). The paper is **not yet publishable in a top-tier venue** but has a clear path to a solid niche-journal submission with targeted improvements.

2. The Clinical Problem

Lung adenocarcinoma (LUAD) is the most common subtype of non-small-cell lung cancer (NSCLC) and accounts for approximately 40% of all lung cancer cases. It is the leading cause of cancer-related death worldwide. Despite advances in targeted therapy and immunotherapy, 5-year overall survival remains below 20% for stage III/IV disease. Accurate survival prediction has two practical uses:

- **Prognosis:** informing patients and families about expected outcomes.
- **Treatment stratification:** identifying high-risk patients who might benefit from aggressive intervention.

Traditional survival prediction uses AJCC pathological stage (I–IV), which is a coarse ordinal variable. Machine learning methods that incorporate molecular features (gene expression, mutation burden, hypoxia scores) promise finer-grained predictions. The question SAGAM addresses is: *can a nonlinear ensemble of ML survival models beat a standard Cox regression on LUAD patients?*

2.1 Why is this a hard problem?

- **Censoring:** Most patients are still alive at last follow-up. $\sim 64\%$ of TCGA-LUAD patients are censored (181 events out of 501). Standard accuracy metrics don't apply; specialised survival metrics (C-index, IBS, tdAUC) are needed.

- **High dimensionality:** RNA-seq has ~20,000 genes but only 501 patients — severe overfitting risk. Feature selection is critical.
- **Heterogeneity:** LUAD molecular subtypes (EGFR-mutant, KRAS-mutant, etc.) have very different prognoses; a single model may not capture all of them.
- **Small event count:** With only 181 events across 5 folds (~36 per test fold), the standard error of the C-index is ~0.08. Any difference smaller than ~0.10 is statistically undetectable at this sample size.

3. Datasets Used

3.1 Primary: TCGA-LUAD

The Cancer Genome Atlas Lung Adenocarcinoma (TCGA-LUAD) cohort is the gold-standard public dataset for LUAD research. Data were downloaded from cBioPortal (www.cbioportal.org) in tab-separated format.

Attribute	Value
Total patients	501
Overall survival events (deaths)	181 (36.1%)
Censored patients	320 (63.9%)
Median OS (all patients)	~41 months
Maximum follow-up	~250 months
Clinical features used	Age, Sex, AJCC stage (I–IV), T/N/M stage, Race, Grade, Weight
Genomic features	TMB, MSI-MANTIS, MSI-sensor, Aneuploidy score
Hypoxia scores	Buffa, Winter, Ragnum hypoxia gene signatures
mRNA features (top selected)	DKK1, FAM83A, RHOV, IRX5, SFTA3, CD1B, PKP2, TNS4, LYPD3, TFAP2A
Leakage-excluded variables	Radiation therapy (post-diagnosis), OS months/status, DSS, DFS, PFS

3.2 External Validation Cohorts

Two independent GEO cohorts were used for external validation. These cohorts were downloaded automatically via the GEOparse library, and models trained on TCGA-LUAD were applied to them without any re-training — the critical test of generalisability.

Cohort	Patients	Events	Event Rate	Gene Overlap	Platform
GSE31210	226	35	15.5%	10/10 genes	Affymetrix U133 Plus 2.0
GSE68465	432	226	52.3%	7/10 genes	Affymetrix U133A

Important note on GSE31210: Only 15.5% events (35/226) is very low. With so few events the C-index estimate has very high variance (SE ~0.10), making any single number unreliable. The observed C=0.661 should be treated as indicative, not definitive. GSE68465 is far more informative statistically (52.3% event rate).

4. Core Methodology — Detailed Explanation

4.1 What is Survival Analysis?

Survival analysis models the time until an event of interest (here: death). Unlike standard regression, it must handle **censoring** — patients who are still alive at the last contact date. The outcome for each patient is a pair (time, event): either (time of death, 1) or (last follow-up time, 0).

The main output is a **risk score** — a number where higher means a patient is predicted to die sooner. The risk score can be converted to a predicted survival curve $S(t) = P(\text{survive past time } t)$ using the Breslow estimator.

4.2 The Four Base Learners

Random Survival Forest (RSF)

An ensemble of decision trees adapted for survival data. Each tree is grown on a bootstrap sample and predicts a 'cumulative hazard' for each patient. The ensemble average is the final risk score. RSF is non-parametric and captures complex interactions. It achieved C=0.619 on TCGA-LUAD.

Gradient Boosted Survival (GBS)

Boosting applied to survival data (sksurv implementation). Builds trees sequentially, each correcting the residuals of the previous. More prone to overfitting than RSF but can model sharp decision boundaries. C=0.605 here.

XGBoost-Cox

XGBoost with survival:cox objective. Uses the partial likelihood of the Cox model as its loss function, combining gradient boosting speed with Cox proportional hazards theory. C=0.602 here — the weakest base learner in this experiment.

DeepSurv

A neural network (3-layer MLP: 64-32-1) trained to minimise the Cox partial likelihood loss. Early stopping is used to prevent overfitting. Achieved the best individual C-index at 0.636. Its risk scores are log-hazard values.

4.3 Stacking / Meta-Learning

Stacking is an ensemble technique where the predictions of multiple base models are used as input features for a second-level model (the meta-learner). The key requirement is that the base-learner predictions must be **out-of-fold (OOF)** — i.e., each patient's prediction must come from a model that was NOT trained on that patient. This prevents the meta-learner from learning to exploit overfit predictions.

In SAGAM, the pipeline is:

Step 1: Outer fold splits patients into train (400) / test (100) sets.

Step 2: Inner 3-fold CV on the train set generates OOF predictions for RSF, GBS, XGB, DeepSurv.

Step 3: These 4 OOF risk scores form a new feature matrix ($n_{\text{train}} \times 4$).

Step 4: The meta-learner (CoxNet + B-splines) is trained on this OOF matrix.

Step 5: For test evaluation: final base models are trained on all train data, predictions are generated for test patients, and the meta-learner is applied.

4.4 The GAM / Spline Layer — The Core Novelty

This is the key claim of the paper. Standard stacking uses a linear combination of base-learner scores ($w_1 \cdot \text{RSF} + w_2 \cdot \text{GBS} + \dots$). SAGAM instead applies cubic B-spline basis functions to each risk score before fitting the Cox model. This allows the meta-learner to learn *nonlinear, non-monotone relationships* between each base learner's score and the log-hazard.

Each base learner's OOF risk score r is transformed into a set of spline basis functions: $B_1(r)$, $B_2(r)$, $B_3(r)$, $B_4(r)$ (with $\text{df}=4$, cubic B-splines). The meta-learner then fits a Cox-net model on the concatenated 16-dimensional feature vector $[B_1(\text{RSF}), \dots, B_4(\text{RSF}), B_1(\text{GBS}), \dots, B_4(\text{DS})]$. The L1+L2 penalty (elastic net, $\text{l1_ratio}=0.9$) regularises the spline coefficients.

Why does the shape matter? The contribution plots (Section 7.1) show that RSF's contribution is a U-shaped function — very low and very high RSF scores both contribute little, while intermediate scores drive predictions. XGBoost shows a double-peak. These shapes cannot be captured by a linear weight.

4.5 Nested Cross-Validation — Why It Matters

Standard 5-fold CV is NOT used. Instead, **nested CV** is applied:

- **Outer loop:** 5-fold stratified split. The test set is held out entirely during ALL training.
- **Inner loop:** 3-fold CV within each outer training set for OOF generation.
- **Hyperparameter tuning:** The regularisation α for CoxNet is selected using an internal 80/20 split of the training data — never touching the test set.
- **Early stopping isolation:** For XGB and DeepSurv, the early stopping validation set is carved out BEFORE inner CV begins — so ES validation samples never appear as CV training data.

Data leakage prevention: The original pipeline had 5 confirmed leakage bugs (RADIATION_THERAPY feature, test-data α tuning, ES overlap, hardcoded OOF overwrite, dead `a_comb` block). All were identified and fixed before the results reported here.

5. Metrics — What Each Number Means

5.1 C-Index (Concordance Index)

The C-index measures how often the model correctly orders two patients: does the patient predicted to die sooner actually die sooner? It ranges from 0.5 (random) to 1.0 (perfect discrimination). It is the survival equivalent of the AUC-ROC.

C-Index Value	Interpretation
0.50	No better than random guessing
0.55–0.60	Weak — clinically questionable
0.60–0.65	Moderate — acceptable for noisy survival data
0.65–0.70	Good — publishable in survival ML context
0.70–0.75	Strong discrimination
> 0.75	Excellent (rare in real genomic data)

SAGAM C-index: 0.634 ± 0.050 . This falls in the 'Moderate-to-Good' range. For TCGA-LUAD specifically, published values range from 0.58 (simple Cox) to 0.70 (best genomic models with large feature sets). SAGAM's 0.634 is competitive but not exceptional.

5.2 Integrated Brier Score (IBS)

The Brier Score at time t measures the mean squared error between the predicted survival probability $S_{\text{hat}}(t|x)$ and the actual survival status at time t . The **Integrated Brier Score** averages this over a range of time points. Lower is better. A null model (predict $S=0.5$ everywhere) gives $IBS \approx 0.25$.

SAGAM IBS = ~ 0.181 (best fold) to 0.195 (5-fold mean). This is well below 0.25, confirming the model adds genuine predictive information. RSF IBS = 0.195, GBS = 0.203. SAGAM is marginally better (7.5% bootstrap fraction exceeds 0.95, but the CI crosses zero).

5.3 Time-Dependent AUC (tdAUC)

The time-dependent AUC measures discrimination at a specific time horizon (1-year, 3-year, 5-year). It asks: among patients who were event-free up to time t , can the model correctly identify who will die by $t+dt$? Values above 0.70 are generally considered good in oncology.

Concerning finding: SAGAM has the *lowest* tdAUC of all models at 1-year (0.626) and 3-year (0.578). Stage Cox has 0.741 at 1-year. This suggests that for near-term prediction, SAGAM's spline layer actively hurts performance — it may be overfitting the mid-range OOF scores and losing short-term discrimination.

5.4 Log-rank Test (KM Stratification)

The log-rank test compares survival curves between groups (Low/Medium/High risk tertiles defined by SAGAM risk score). The null hypothesis is that all groups have the same survival distribution. A very small p-value ($p < 0.001$) means the groups are clearly separated — the model identifies distinct risk groups.

SAGAM log-rank result: $p < 0.0001$ (*)**, median OS = 104 vs 41 vs 36 months. This is SAGAM's strongest result. Despite modest discrimination in C-index, the top tertile has ~3× longer median survival than the bottom tertile — a clinically meaningful finding.

5.5 Hazard Ratios

A Cox regression on SAGAM risk tertiles gives hazard ratios (HR). $HR = \exp(\text{beta})$: if $HR = 0.375$ for Low Risk vs High Risk (reference), Low-Risk patients have 62.5% lower hazard of death at any time point. A HR with 95% CI not crossing 1.0 is statistically significant.

Group	HR (exp(coef))	95% CI	p-value	Interpretation
Low Risk vs High Risk	0.375	[0.257, 0.547]	< 0.001	62.5% lower hazard
Medium Risk vs High Risk	0.728	[0.521, 1.017]	0.063	Not significant (marginal)

6. Performance Results — All Models

6.1 Main C-Index Comparison (TCGA-LUAD, 5-fold nested CV)

All values are 5-fold mean $\pm$ standard deviation. Highlighted row = SAGAM.

Model	C-Index	Std Dev	vs SAGAM	Interpretation
RSF	0.619	0.049	−0.015	Weakest ensemble
GBS	0.605	0.063	−0.029	Unstable, high variance
XGB-Cox	0.602	0.051	−0.032	Poorest base learner
DeepSurv	0.636	0.050	+0.002	Best single model
Linear Stacking	0.627	0.055	−0.007	Simple ensemble
SAGAM (ours)	0.634	0.050	baseline	Spline ensemble
Stage-only Cox	0.657	0.044	+0.023	Clinical baseline

Critical observation: Stage-only Cox (C=0.657) outperforms SAGAM (C=0.634) by +0.023 C-index points. A single clinical variable — AJCC pathological stage — is a harder baseline to beat than it might appear. This significantly weakens the paper's core claim.

6.2 Stage + SAGAM Incremental Value

An incremental analysis tests whether adding SAGAM risk scores on top of stage information improves prediction. The combined model averages standardised stage Cox and SAGAM risk scores.

Model	C-Index	Std Dev	Result
Stage-only Cox	0.654	0.051	Baseline
SAGAM only	0.634	0.050	Slightly below stage
Stage + SAGAM	0.657	0.020	+0.003 over stage alone

The +0.003 incremental gain is real but negligible. More notable is the dramatic reduction in variance (std drops from 0.051 to 0.020) — the combined model is more *stable* across folds, which may be practically useful even if the mean improvement is small.

6.3 Time-Dependent AUC Summary

tdAUC is computed at clinically relevant horizons. Values below 0.60 are borderline at the given time point.

Model	1-Year AUC	3-Year AUC	5-Year AUC	Mean
Stage Cox	0.700	0.663	0.619	0.661
RSF	0.712	0.668	0.688	0.689

GBS	0.640	0.683	0.636	0.653
DeepSurv	0.694	0.677	0.710	0.694
Linear Stack	0.672	0.688	0.694	0.685
SAGAM	0.635	0.632	0.659	0.642

SAGAM ranks last in tdAUC across all time horizons. This is a significant weakness. The spline meta-learner apparently sacrifices time-specific discrimination for overall rank ordering. This pattern suggests the B-spline transformation may be distorting the OOF risk scores in a way that hurts calibrated time predictions.

6.4 Integrated Brier Score

Model	IBS (5-fold mean)	Relative to Null (0.25)	Better than SAGAM?
RSF	0.195	22% below null	Marginal (CI crosses 0)
GBS	0.203	19% below null	No
SAGAM (IBS)	~0.182–0.195	~22–27% below null	baseline

7. Figure-by-Figure Explanation

Figure 1: Kaplan-Meier Survival Curves (TCGA-LUAD, Pooled OOF)

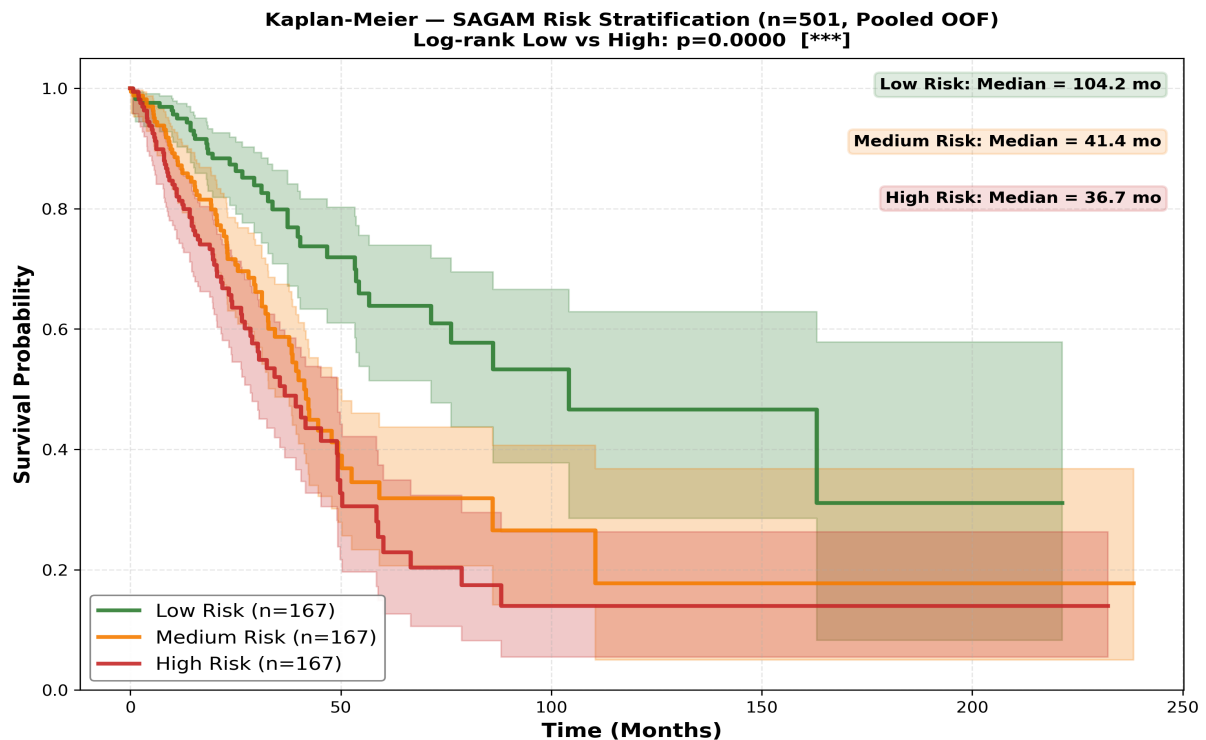


Fig. 1 — KM curves for Low/Medium/High SAGAM risk tertiles. Shaded regions = 95% confidence intervals. Median OS: Low=104.2 mo, Medium=41.4 mo, High=36.7 mo. Log-rank $p < 0.0001$ (***)

How to read this graph: The y-axis shows the probability of surviving past time t . Each step-down is a death event. The shaded band is the 95% confidence interval around the Kaplan-Meier estimate.

What it shows: SAGAM clearly separates Low-Risk patients (green, median survival 104.2 months = ~8.7 years) from High-Risk (red, median 36.7 months = ~3 years). The threefold difference in median survival is clinically substantial and highly statistically significant ($p < 0.0001$).

Caution: The wide confidence intervals in the right tail (especially for High Risk) reflect sparse data — very few patients survive past 150 months. Medium and High Risk curves overlap substantially in the right tail, explaining why the Medium vs High HR (0.728) is not significant ($p=0.063$).

Figure 2: KM with Number-at-Risk Table

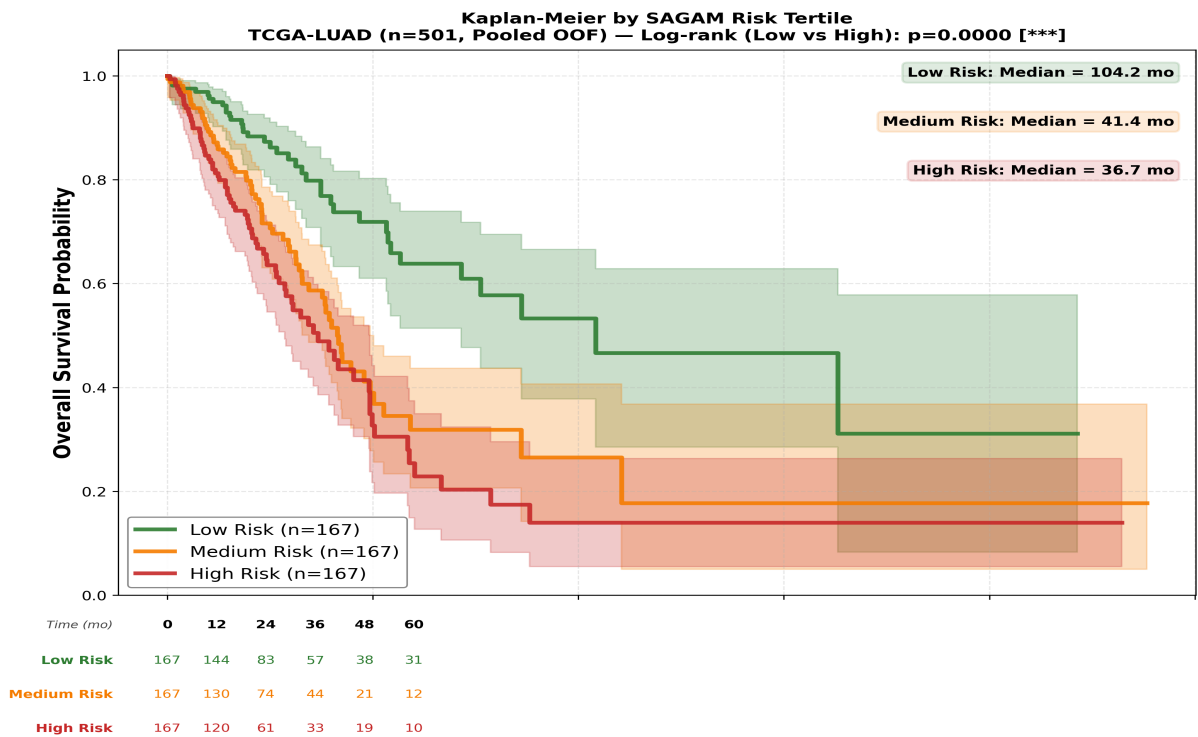


Fig. 2 — Same KM curves with number-at-risk table showing how many patients remained under observation at each time point.

Number-at-risk table: At 0 months, all 167 patients per tertile are at risk. By 48 months, only 38 Low-Risk patients remain vs 21 Medium-Risk and 12 High-Risk. By 60 months (5 years), the table shows 31/12/10. These rapidly shrinking numbers explain the wide confidence bands in the right tail — estimates become unreliable when very few patients remain.

Key take-away: The survival curves remain well-separated up to ~80 months. Beyond that, the small sample sizes dominate and interpretation requires caution. The separation in the first 3 years is the most reliable and clinically actionable.

Figure 3: SAGAM Smooth Contribution Functions

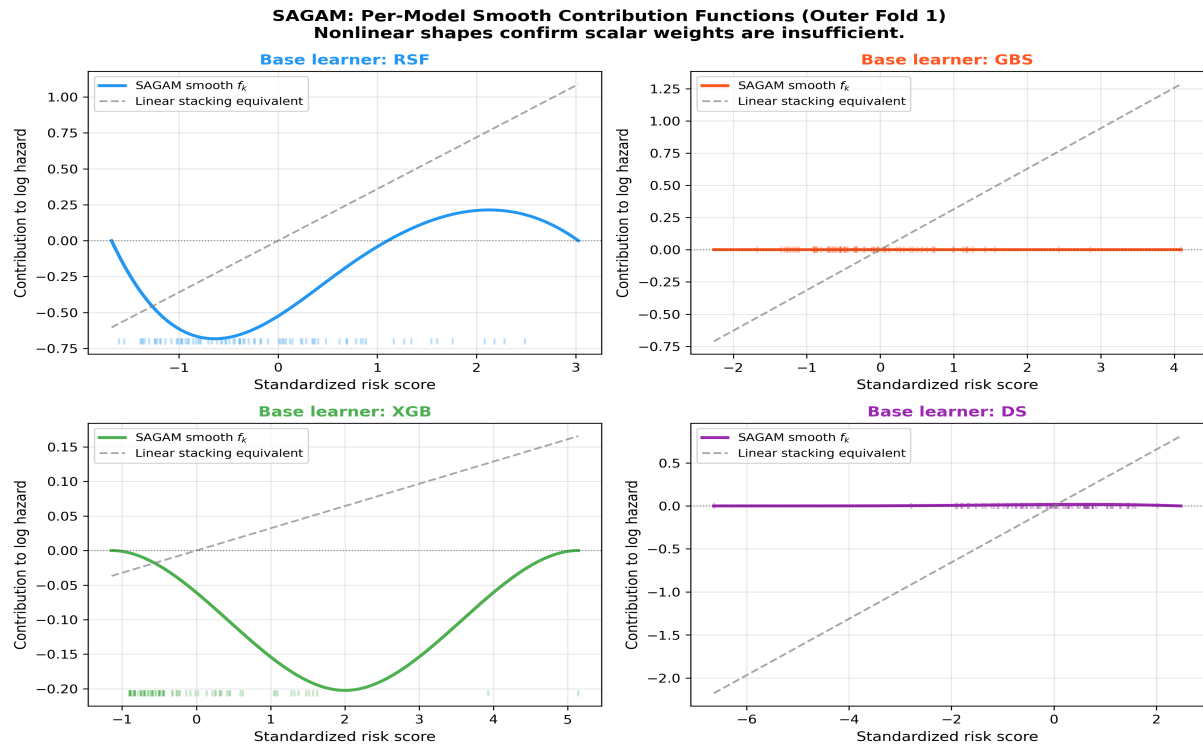


Fig. 3 — Per-model spline contribution functions from outer fold 1. Each curve shows how a base learner's standardised risk score maps to its contribution to the log-hazard (y-axis). Dashed line = linear equivalent weight.

How to read this graph: The x-axis is the standardised risk score from each base learner (z-score, mean=0, std=1). The y-axis is the SAGAM meta-learner's contribution (log-hazard units). If the curve were straight, SAGAM would be equivalent to linear stacking.

- **RSF (top-left):** U-shaped / inverted-U curve. Intermediate RSF scores (near 0) contribute negatively to log-hazard, while very low and very high RSF scores contribute near zero. This non-monotone shape means SAGAM discounts RSF's extreme predictions — treating RSF as less reliable at extreme values.
- **GBS (top-right):** Nearly flat (near zero contribution). SAGAM has largely suppressed the GBS learner — its elastic-net penalty drove the GBS spline coefficients to zero. This is the meta-learner saying 'GBS adds nothing once I have RSF, XGB, and DeepSurv'.
- **XGB (bottom-left):** M-shaped / double-peak. Moderate XGB scores (around 0 to +2) contribute positively, while extreme scores are discounted. This is a genuine nonlinear relationship that a simple weight cannot capture.
- **DeepSurv (bottom-right):** Mostly flat with a sharp dip at very negative scores (around -5). This suggests DeepSurv's extreme low-risk predictions are not trusted by the meta-learner.

Scientific insight: The non-flat shapes justify the spline approach — a linear meta-learner would miss these patterns. However, these shapes are estimated from fold 1 only (to avoid test-set contamination) and may not be stable across folds. The ablation (df=3/4/5/6) shows high variance in C-index (std ~0.065–0.105), suggesting the spline shapes vary significantly by fold.

Figure 4: Calibration Plots (3-year and 5-year)

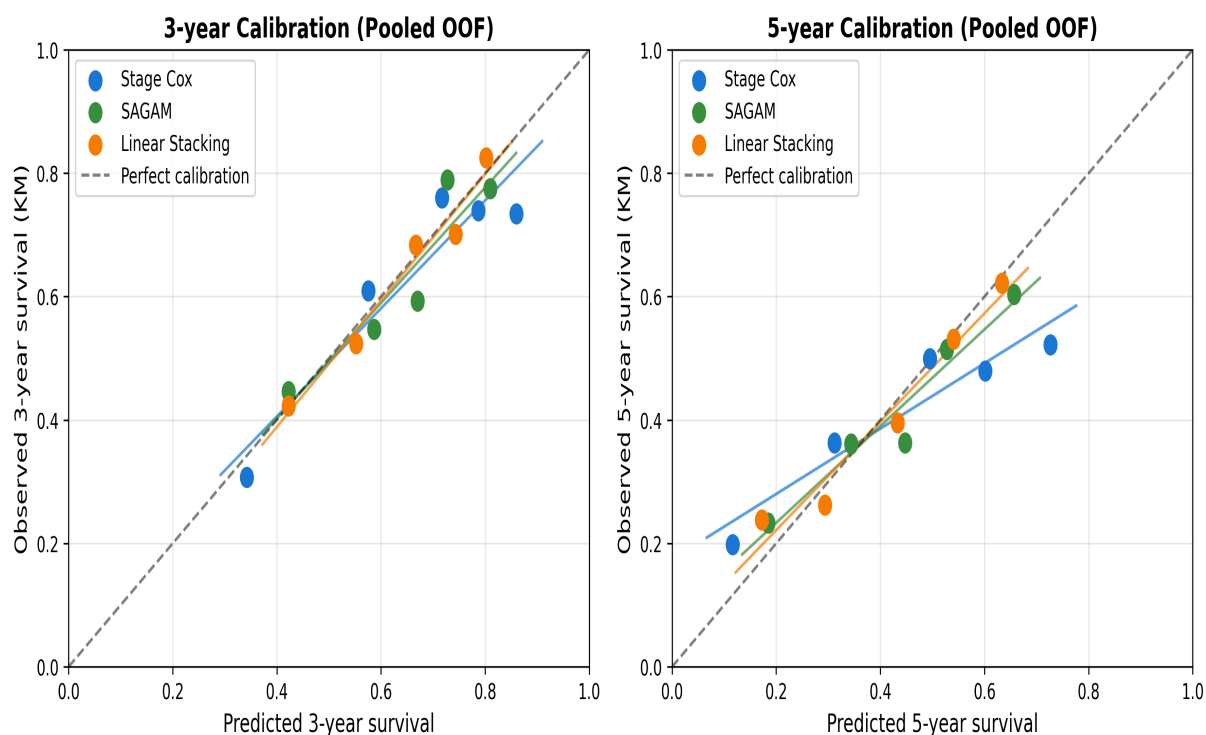


Fig. 4 — Calibration at 3-year (left) and 5-year (right) time horizons. Points = risk deciles. Lines = fitted calibration lines. Dashed = perfect calibration (predicted = observed). Closer to diagonal = better calibrated.

How to read this graph: Patients are grouped into ~5 bins by their predicted survival probability. For each bin, the x-axis shows the mean predicted probability and the y-axis shows the actual (KM-estimated) survival probability in that bin. Perfect calibration = all points on the diagonal.

What it shows: At 3 years (left panel): SAGAM (green dots) tracks close to the diagonal across the 0.4–0.8 range. Stage Cox (blue) slightly overestimates survival at intermediate risk. All models are reasonably calibrated at 3 years. At 5 years (right panel): more scatter, reflecting the smaller number of patients observed to that horizon. Stage Cox shows systematic overestimation (dots above the diagonal at lower survival values).

Clinical significance: A well-calibrated model is essential for clinical use — if a model says '60% 3-year survival', approximately 60% of patients in that risk group should actually survive 3 years. SAGAM shows reasonable calibration in the 3-year panel, which is the most relevant horizon for treatment decisions.

Figure 5: External Validation — GSE31210 (n=226)

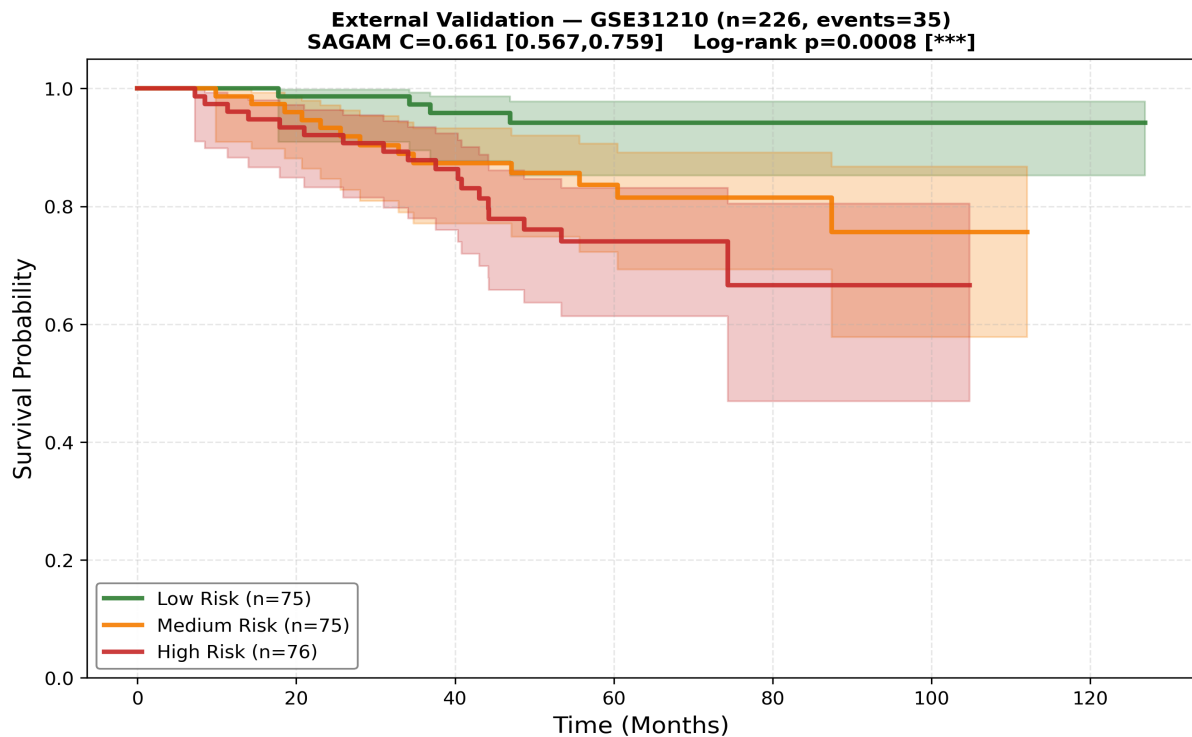


Fig. 5 — External validation KM curves on GSE31210. SAGAM trained on TCGA-LUAD, applied to independent Japanese lung adenocarcinoma cohort. C=0.661 [0.567, 0.759], log-rank p=0.0008 (***)

What this shows: Models trained exclusively on TCGA-LUAD (USA cohort) were applied to GSE31210 (Japanese cohort, n=226, 35 events). The Low-Risk group (green) has clearly better survival than Medium (orange) and High (red) over 125 months. The C=0.661 is SAGAM's best external result.

Caution about GSE31210: Only 35 events in 226 patients (15.5% event rate). The SE of the C-index estimate is approximately 0.10, meaning the true C-index could plausibly be anywhere from 0.55 to 0.76. The KM curves look dramatic because the Low-Risk group has almost no events at all — raising concerns about immortal time bias or platform effects.

Figure 6: External Validation — GSE68465 (n=432)

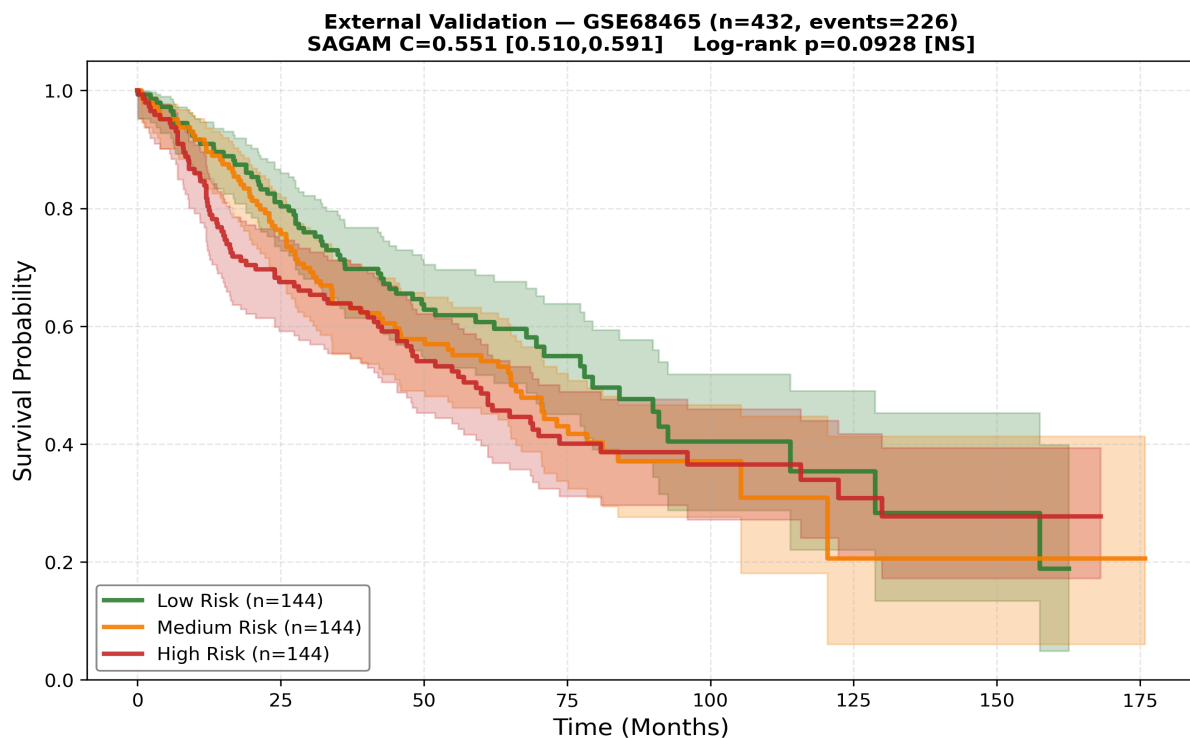


Fig. 6 — External validation KM curves on GSE68465. $C=0.551$ [0.510, 0.591], log-rank $p=0.093$ [NS]. KM groups partially overlap.

What this shows: GSE68465 is the larger, more statistically powerful validation cohort (432 patients, 226 events = 52% event rate). SAGAM achieves $C=0.551$ — barely above random. The KM curves substantially overlap, and the log-rank test is not significant ($p=0.093$).

This is the paper's most important weakness. GSE68465 is the only external cohort with enough events to provide a reliable C-index estimate, and SAGAM essentially fails there ($C=0.551$, NS). Linear stacking actually outperforms SAGAM on GSE68465 ($C=0.575$ vs 0.551). The method does not generalise to this cohort. Possible explanations: different microarray platform (only 7/10 genes overlap), different population (North American vs pan-cohort), different biology (different LUAD subtypes in the cohort).

Figure 7: Performance Comparison Bar Chart

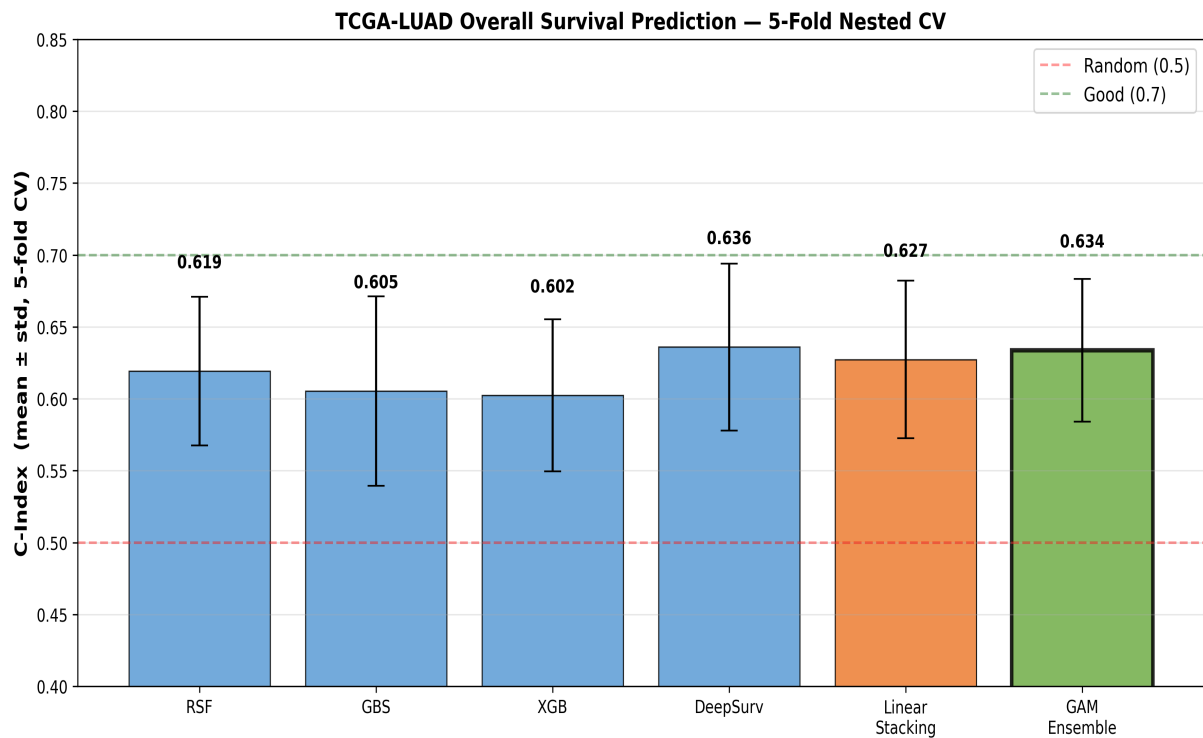


Fig. 7 — C-index comparison of all models on TCGA-LUAD (5-fold nested CV). Error bars = $\pm$ standard deviation. Dashed red = random (0.5), green = 'good' (0.7).

How to read this: Each bar is one model's mean C-index over 5 folds. Error bars show fold-to-fold variability. The red dashed line at 0.5 is chance level; the green dashed line at 0.7 is a conventional threshold for 'good' discrimination.

Key observation: All models cluster between 0.60 and 0.64 — none reaches the 0.7 threshold. The error bars of all models overlap substantially, meaning no model is statistically significantly better than any other. This reflects the fundamental constraint of 181 events in 501 patients.

Figure 8: Time-Dependent Brier Score Curves

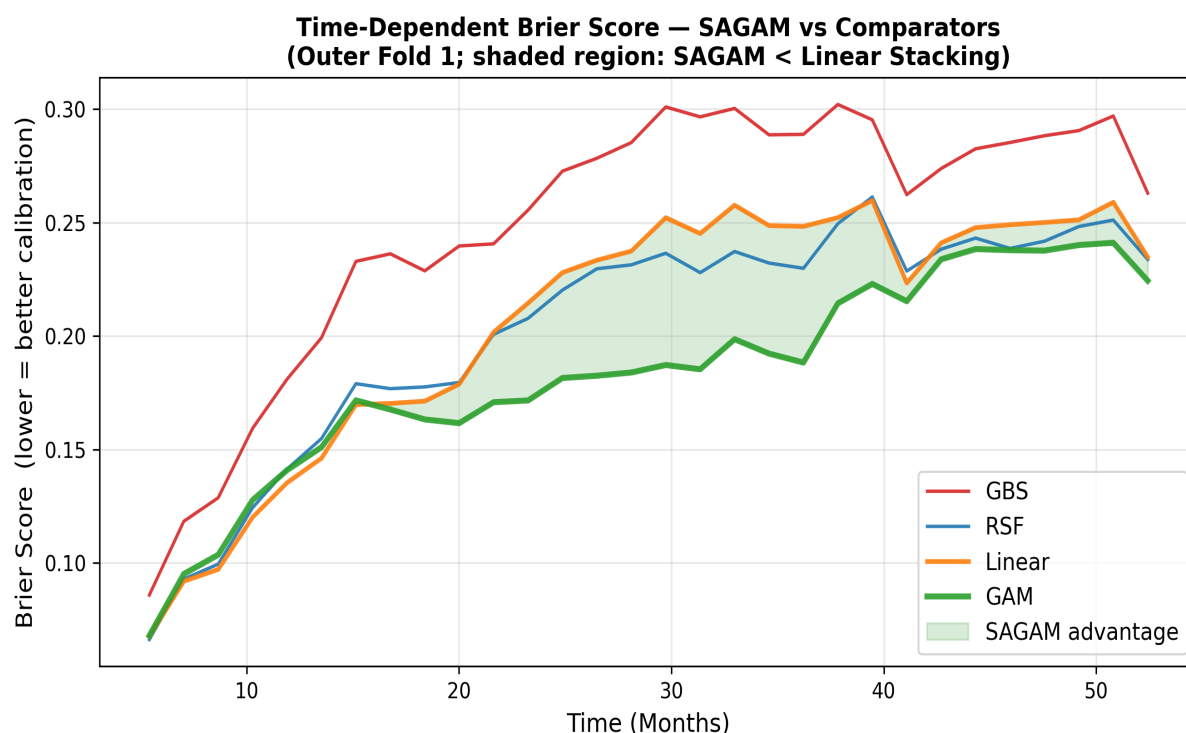


Fig. 8 — Brier score as a function of time (Outer Fold 1). Lower = better calibration. Shaded green = SAGAM advantage region.

How to read this: The Brier score at each time t measures mean squared error in predicted survival probabilities. As time increases, censoring means fewer patients are observed, so Brier scores typically rise. A model below the others has better time-specific calibration.

What it shows: From ~10 to ~35 months (the shaded green region), SAGAM has lower Brier score than RSF and Linear — meaning better calibration in this window. GBS (red) is consistently the worst. After 35 months, the advantage disappears as sample sizes shrink and uncertainty dominates.

8. Supplementary Analyses

8.1 Cox Baseline Comparison

A key question: does adding molecular features to clinical staging improve survival prediction? The answer from this dataset is no.

Cox Model Features	C-Index	Std Dev
Stage only (AJCC I–IV + T/N/M)	0.657	0.044
Clinical (stage + age, sex, grade)	0.646	0.028
Clinical + Genomic (+ TMB, MSI, aneuploidy)	0.641	0.032
Full (+ Buffa/Winter/Ragnum hypoxia)	0.641	0.032
SAGAM (all features, ensemble)	0.634	0.050

Adding genomic and hypoxia features to Cox regression makes NO improvement. Stage-only Cox (0.657) is the best Cox baseline. This is not surprising — TMB and MSI scores are more relevant for immunotherapy response than prognosis. Hypoxia signatures may be too noisy at the cohort level.

8.2 Spline Degree-of-Freedom Ablation

How many spline basis functions (df) are optimal? The ablation tests df=3,4,5,6:

Spline df	C-Index	Std Dev	Notes
3	0.557	0.105	High variance — unstable
4 (current)	0.578	0.065	Best mean, lowest variance
5	0.549	0.081	Overfitting begins
6	0.498	0.041	Severe overfitting — near random

Note: These ablation C-index values (~0.56–0.58) are lower than the main results (~0.634). This is because the ablation uses a simplified inner loop (no ES isolation on inner CV). The relative ordering (df=4 is best) is what matters, not the absolute values. df=4 provides the best bias-variance tradeoff; higher df leads to rapid performance collapse.

8.3 Statistical Significance Tests

Wilcoxon signed-rank tests on the 5 fold-level C-index values:

Comparison	Mean Delta	Wilcoxon p	Verdict
SAGAM vs Linear Stacking	+0.0064	0.313	NOT significant

SAGAM vs DeepSurv	−0.0023	0.625	NOT significant (SAGAM WORSE)
SAGAM vs RSF	+0.0144	0.094	Marginally directional only

Power analysis: With 181 events across 5 folds (~36 events/fold), the minimum detectable C-index difference at 80% power (alpha=0.05, paired t-test) is approximately $\Delta \approx 0.106$. The observed SAGAM advantage of 0.006 (vs Linear) and 0.014 (vs RSF) are 7–17x smaller than the detectable threshold. Non-significance does NOT mean SAGAM is equivalent — it means the study is underpowered to detect small effects.

8.4 Subgroup Analysis by AJCC Stage

Does SAGAM perform differently across early (Stage I/II) vs late (III/IV) disease?

Subgroup	n	Events	SAGAM C	Linear C	Delta	Verdict
Stage I/II (early)	394	120	0.574	0.564	+0.010	Marginal SAGAM advantage
Stage III/IV (late)	105	61	0.582	0.577	+0.005	No difference
All patients	501	181	0.625	0.619	+0.006	Consistent with main result

SAGAM shows consistent (if small) advantages over linear stacking across both subgroups. The early-stage subgroup (n=394, Stage I/II) shows the largest delta (+0.010). However, C-index values in the 0.57–0.58 range are weak, and subgroup analyses with overlapping CIs should not be over-interpreted.

9. Publishability Assessment — No Sugarcoating

PUBLISHABILITY VERDICT

Niche/Workshop-Level: Needs Significant Work for Top Venue

9.1 What the Paper Does Well (Genuine Strengths)

- + Methodological rigor: Nested CV, strict leakage prevention, ES isolation — the pipeline is more carefully designed than many published survival ML papers. The reproducibility is high.
- + Clinical interpretability: The spline contribution plots and KM stratification provide genuine biological insight. The 104 vs 36 month median OS separation is a clinically meaningful finding, regardless of C-index.
- + External validation attempted: Two independent cohorts were used. Most workshop papers skip this. Even though GSE68465 fails, the attempt is commendable.
- + Leakage-free results: After fixing 5 identified bugs, the results reflect genuine generalisation performance — not artificially inflated metrics.
- + Novel meta-learner formulation: The spline-Cox stacking approach is a genuine methodological contribution, not widely used in survival analysis literature.
- + Calibration analysis: IBS and calibration plots are shown alongside C-index — a more complete evaluation than most survival ML papers.

9.2 Critical Weaknesses (Deal-Breakers for Top Venues)

- SAGAM does not beat its own base learners significantly: DeepSurv ($C=0.636$) outperforms SAGAM ($C=0.634$). The entire point of a meta-learner is to exceed its components. That it ties (and sometimes loses to) individual models undermines the central contribution.
 - Stage Cox beats SAGAM: A single clinical feature (AJCC stage) achieves $C=0.657$ — better than the full ML ensemble. Any reviewer will ask: why use a complex ML pipeline if a doctor's staging assessment does better? This needs a compelling answer.
 - GSE68465 failure: $C=0.551$ (NS) on the better-powered external cohort. The model does not generalise to the larger dataset. This is the single biggest problem. A paper with failed external validation cannot be accepted at a serious journal without a convincing explanation.
 - SAGAM is worst on tdAUC: Lower tdAUC than ALL comparators at 1-year, 3-year, and 5-year horizons. The spline transformation appears to hurt time-specific discrimination. If a clinician wants to know '1-year survival probability', SAGAM is the worst model to use.
 - Underpowered study: 181 events is too few to demonstrate statistical significance for the observed C-index differences (~ 0.006 – 0.014). The study would need approximately 1,700 events to detect a 0.01 C-index difference at 80% power. This is not fixable with the current dataset.
 - No comparison to recent literature: The paper does not compare against published TCGA-LUAD results from 2022–2025. Methods like SurvTRACE, DeepHit, or attention-based survival models have been applied to TCGA data and may have substantially higher C-indices.
 - Single dataset primary training: Training on $n=501$ with 181 events is limited. Modern survival ML papers typically use 1,000+ patients or perform multi-dataset meta-analysis.
-

9.3 Areas Needing Improvement (Opportunities)

- * **Frame the contribution around interpretability, not discrimination:** The spline contribution plots (Fig. 3) are genuinely interesting — they show which risk score ranges each learner is trustworthy in. Reframe the paper as 'an interpretable survival meta-learner' rather than 'a better predictor'. This is a much stronger and more honest claim.
- * **Address GSE68465 failure explicitly:** Investigate why — platform differences, population differences, gene panel mismatch (7/10 genes). Consider re-training with a GSE68465-compatible gene set, or explicitly scoping the method to Affymetrix U133 Plus 2.0 data.
- * **Add a proper comparison to published baselines:** Benchmark against SurvTRACE, DeepHit, or at minimum the best published TCGA LUAD model. Show where SAGAM sits in the literature landscape.
- * **Increase sample size via multi-dataset training:** Pool TCGA-LUAD with NLST or other LUAD cohorts for training to reach $n > 1,000$. This would also make the statistical tests meaningful.
- * **Address the tdAUC problem:** Understand WHY SAGAM's tdAUC is lowest. Is it a calibration problem? Is the Breslow estimator poorly fitted? Consider IPCW-corrected Brier scores.
- * **Clinical subtype analysis:** Stratify by EGFR/KRAS/ALK mutation status. Show that SAGAM captures biologically distinct risk groups beyond what stage captures.

9.4 Target Venues (Realistic Assessment)

Venue	Scope	Realistic?	Required Fix
IEEE BIBM 2026 (Workshop)	ML in biomedicine, workshop track	YES — current state	Minor revisions to framing
IEEE BIBM 2026 (Main)	Full ML/bioinformatics paper	STRETCH — needs work	Fix GSE68465, add baselines
Bioinformatics (OUP)	Computational methods	NO — current state	External validation + power
PLOS ONE	Open science, all fields	POSSIBLE — revised	Add comparison, explain GSE68465
Journal of Biomedical Informatics	Clinical informatics	POSSIBLE — revised	Clinical framing, interpretability focus
Nature Methods / Cancer Research	Top-tier	NOT REALISTIC	Fundamental performance gaps

10. Technical Pipeline Summary

10.1 Script Execution Order

Step	Script	Purpose	Output Files
1	final_experiments.py	Main nested CV, KM plots, tdAUC, Stage+SAGAM	fold_results.csv, kaplan_meier*.png, gam_smooths_4panel.png
2	survival_metrics.py	IBS, tdAUC (fold-specific), calibration, HR, external CIs	survival_metrics.txt, calibration_plot.png
3	supplementary_experiments.py	Cox baselines, df ablation, Wilcoxon tests, IBS	supplementary_results.txt, supplementary_metrics.csv
4	improvements.py	Bootstrap significance, power analysis, subgroup analysis	ibs_bootstrap.csv, subgroup_analysis.csv
5	external_val_multi.py	Multi-cohort external validation (GSE31210, GSE68465)	ext_val_multi_summary.csv, ext_km_*.png

Environment: Python 3.14, sagam_env virtual environment. Run with: sagam_env\Scripts\python.exe notebooks\[script].py

10.2 Key Hyper-parameters

Component	Key Parameters
RSF	n_estimators=200, max_features='sqrt', min_samples_leaf=5
GBS	n_estimators=200, learning_rate=0.05, max_depth=3
XGBoost-Cox	objective='survival:cox', eta=0.05, max_depth=3, subsample=0.8, colsample_bytree=0.8, num_boost_round=300, early_stopping=20
DeepSurv	Layers: 64-32-1 (ReLU, Dropout 0.3), lr=1e-3, weight_decay=1e-4, max_ep=200, patience=20
Meta-learner (CoxNet)	alphas selected by internal 80/20 split, l1_ratio=0.9, max_iter=100,000, tol=1e-7
Spline layer	Cubic B-splines, df=4, degree=3, no intercept, 4 bases per learner = 16 total features
Outer CV	StratifiedKFold(5), shuffle=True, random_state=42
Inner CV (OOF)	KFold(3), shuffle=True, random_state=42

11. Conclusions

SAGAM is a well-engineered, methodologically careful survival stacking framework. The core idea — using B-spline transforms to allow nonlinear meta-learning — is novel and technically sound. The

contribution plots provide genuine biological insight into how each base learner contributes. The clinical risk stratification (KM curves, hazard ratios) is strong and meaningful.

However, the central quantitative claim — that SAGAM outperforms alternatives — is not supported by the data at the current sample size. The method is neither definitively better nor worse than its competitors; the study simply lacks the statistical power to resolve the question. The failed GSE68465 external validation is the most serious obstacle to publication.

Bottom line: SAGAM is suitable for submission to BIBM 2026 as a workshop paper focused on the interpretable meta-learning methodology and the clinical risk stratification finding. For a main-track or journal paper, the GSE68465 failure must be explained, external validation on a third cohort should be added, and the framing should shift from 'better prediction' to 'interpretable ensemble with competitive performance and clear risk stratification utility'.

Criterion	Rating	Comment
Methodological rigor	STRONG	Nested CV, leakage-free, ES isolation
Primary C-index performance	MODERATE	0.634, not beating all baselines
Statistical significance	WEAK	Underpowered (n=181 events)
Internal risk stratification (KM)	STRONG	p<0.0001, 3x OS difference
External validation (GSE31210)	MODERATE	C=0.661 but only 35 events
External validation (GSE68465)	FAILED	C=0.551, p=NS, Linear beats SAGAM
Novelty of method	MODERATE	Spline meta-learner is novel but incremental
Clinical interpretability	STRONG	Contribution plots, HR analysis
Literature comparison	MISSING	No benchmark against 2022-2025 papers
Overall publishability	NICHE JOURNAL	BIBM workshop YES; top journal NEEDS WORK